AQUA-VES™ Offshore Water Treatment System



SHEAR ENHANCED FILTRATION SYSTEM MINIMIZES WASTE WATER DISPOSAL



The AQUA-VES Offshore Water Treatment System effectively treats offshore slop water so it may be safely discharged overboard. The system can treat slop water that is contaminated with high levels of oil and solids. The membrane filters reject solids and oil to produce a clean water that contains less than 30 mg/L oil and meets the requirements for offshore discharge. The system is completely automated and uses an integrated oil-in-water analyser to ensure the clean water meets the environmental requirements. If the system detects an output near the discharge limits, it will automatically divert the water back to the feed tank for re-processing.

While process rates and efficiencies vary depending on the feed density, suspended solids, dissolved solids, viscosity, and temperature, the AQUA-VES significantly reduces the amount of waste water that must be shipped to shore for disposal. Up to 80% of a typical slop water volume can be discharged after treatment. Thus, only the concentrate, which is as little as 20% of the original volume, would then need to be shipped to shore for disposal. This results in a significant cost savings for the operator. This will also reduce the slop water storage volume required on the rig.

The system is designed to run automatically and to maximize the life of the machine. The PLC has several programmed set-points that will automatically stop the process if any of the measurements reaches an alarm level. This maximizes the life of the membranes that could otherwise be reduced by excessive processing rates and pressures. The automation system also minimizes operator intervention. All process times, volumes, and cleaning cycles can be programmed to perform when desired.



Feed



Clean Filtered Water



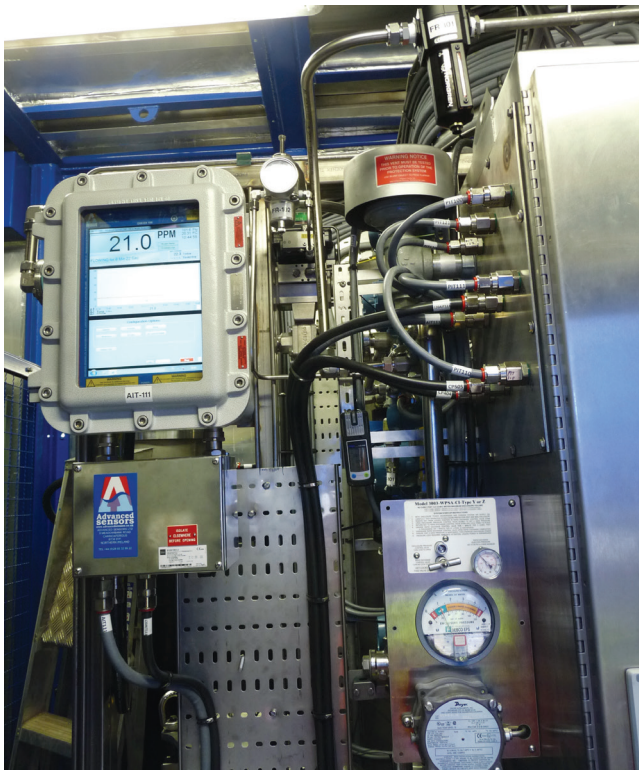
Concentrated Waste

NOV Wellbore Technologies

4310 N Sam Houston Pkwy East
Houston, Texas 77032, USA
Phone: 713 482 0500 • Fax: 713 482 0699

FluidControl@nov.com nov.com/wellsite

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FEATURES	BENEFITS
Fully integrated system	One lift saves time
PLC and HMI controls	Easy to operate (requires only one operator)
Small footprint	Minimizes space requirements on the rig
Easy installation at rig site	Reduces time to setup and become operational
Vibratory enhanced separation	Reduces fouling and increases membrane life
Long membrane life	Lowers operating costs
Reduces waste by up to 80% by volume	Reduces shipping and onshore disposal costs
Integrated oil-in-water analyser	Ensures all discharged water meets regulations

Specifications and Dimensions*

Model	Offshore System
Processing Rates	10-20 bbl/hour (1.5-3 m^3/hour)
Power Requirement	460 V, 60 Hz, 3 phase, 100 A
Dimensions (L x W x H)	155 in x 108 in x 135 in (3925 mm x 2744 mm x 3429 mm)
Max Gross Weight	24,251 lb (11 000 kg)
Certifications	DNV 2.7-1, ATEX Zone 2. CE Ex II 3 G c Ex d e ib m pz IIB T3 Gc
**Approximate specifications. For more information please contact your local NOV WellSite Services representative.*	

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